



Fouling resistant silicone coating with self-healing induced by metal coordination

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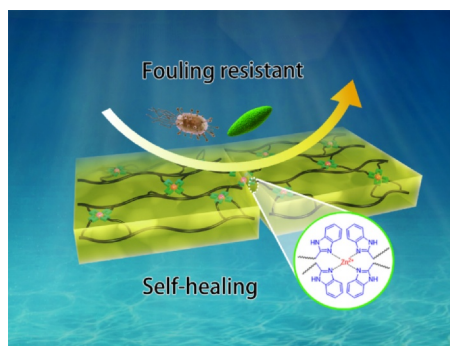
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HIGHLIGHTS

- Novel silicone coating with excellent self-healing property under room temperature induced by metal-coordination.
- High adhesion strength to substrates.
- Excellent fouling resistant performance toward marine bacterial and diatom.

GRAPHICAL ABSTRACT

Silicone coating cross-linked via metal coordination has been prepared, which exhibits excellent self-healing and fouling resistant abilities.



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ABSTRACT

Silicone based fouling release coatings with eco-friendliness have received growing interest. Yet, their poor fouling resistance and self-repairing inability restrict their applications. Here, we report a new poly(dimethylsiloxane) (PDMS) coating cross-linked via coordination bonds between 2-(2-benzimidazolyl)ethanethiol (BET) and zinc ions. Such modified PDMS still has low surface energy but improved adhesion strength in comparison with PDMS elastomer. The metal coordination makes its cross-linking reversible, so it exhibits good self-healing in air or artificial seawater at room temperature. Moreover, the presence of zinc-imidazole complexes with antifouling function significantly improves the fouling resistance against the marine bacteria *Pseudomonas* sp. and the diatom *Navicula incerta*. The study provides an approach to develop high performance antifouling coating.

1. Introduction

Marine biofouling comes from the settlement and accumulation of marine microorganisms, plants, and animals on surfaces immersed in seawater. It not only increases the hydrodynamic drag, fuel consumption and acceleration of corrosion but also introduces alien species into

the ecosystem [1–3]. It has become a worldwide problem today. Various coatings have been used to inhibit marine biofouling for decades [4]. However, most of them contain biocides and have detrimental effects on marine environments [5].

Silicone-based fouling-release coatings (FRCs) have attracted great attention because of their environmentally benign, fouling-release, and

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drag-reducing properties [6]. FRCs are generally based on poly(dimethylsiloxane) (PDMS) elastomer due to its low surface energy, high elasticity and low surface roughness [2]. Marine organisms weakly attach to the surfaces and can be easily removed by shear forces of water flow or gentle mechanical cleaning [7,8]. Also, the smooth surfaces can effectively reduce the drag during navigation. Yet, they do not have fouling resistant ability and generally exhibit poor antifouling performance under static conditions. To improve the fouling resistance of FRCs, amphiphiles [9–12], zwitterionic polymers [13–15], quaternary ammonium salts [16,17] and antifoulants [18,19] were introduced to PDMS elastomer. However, the modification usually leads a decline in the mechanical properties. So far, most of them have poor adhesion to substrates without tie coat. Besides, they suffered from mechanical damage due to its low modulus. Generally, mechanical damage can frequently happen on the silicone elastomer due to cutting, tearing, and puncturing. In particular, mechanical damage will result in local unprotected areas where barnacles can strongly adhere on it. Therefore, specific touch-ups in dry-dock are needed to overcome the mechanical defects, which greatly reduce the service time and increase the cost [20].

Self-healing polymers via reversible cross-links, such as dynamic covalent bonds [21–23], metal coordination bonds [24–28], and hydrogen bonds [29–31] have been developed rapidly for utilization in coatings, soft robots and wearable devices [32–34]. Functionalized PDMS elastomer with self-healing property is expected to improve the durability of antifouling coatings. To date, only a few reports deal with self-healing coatings for marine antifouling [35–38]. Yet, they usually exhibit poor static antifouling performance. Some organic antifoulants or silicone oil had to be introduced, which probably have adverse effects on marine ecosystem [39]. It is a challenge to achieve good antifouling ability without addition of the chemicals.

In this study, we have developed PDMS-based coating with good antifouling performance and self-healing ability, where benzimidazole groups attached on PDMS chains form complexes with zinc ions through metal coordination, acting as both reversible cross-linkers and non-leaching antifouling agents, just as the proverb says “Kill two birds with one stone”. Such complexes with antifouling function significantly improve the silicone coating’s fouling resistance. The synergistic effect of dynamic metal-imidazole complexes and hydrogen bonding makes the coating has not only good self-healing ability but also high adhesion strength. We have examined the wettability, adhesion and antifouling properties of the coating, and correlated them with its structure and composition. Our aim is to develop a new and innovative strategy to prepare high performance self-healing antifouling coating.

2. Materials and methods

2.1. Materials

Octamethylcycloterasiloxane (D4), teramethylteravinylcycloterasiloxane (D4^{Vi}), hexamethyldisiloxane (MM), teramethylammonium hydroxide ((CH₃)₄NOH, 10 wt% in methanol), 3-mercaptopropionic acid, 1,2-diaminobenzene, 2,2-dimethoxy-2-phenylacetophenone (DMPA), and zinc chloride (ZnCl₂) were purchased from Aladdin (China) and used without further purification. PDMS elastomer (Sylgard 184) was purchased from Dow Corning. Tetrahydrofuran (THF) and N,N-dimethyl formamide (DMF) were dried over calcium hydride before distillation. Artificial seawater (ASW) was prepared according to standard ASTM D1141-98 (2013).

2.2. Synthesis of 2-(2-benzimidazolyl)ethanethiol (BET)

BET was synthesized according to the literature [40]. 3-Mercaptopropionic acid (7.83 g, 73.5 mmol) and 1,2-diaminobenzene (6.57 g, 60.8 mmol) were added to 30 mL of a 4 M HCl aqueous solution, and the mixture was refluxed at 100 °C under argon atmosphere for 40 h

(Scheme 1a). After the reaction, 470 mL of deionized water was added to the mixture to obtain a total volume of 500 mL. Then, a certain volume of 4 M NaOH aqueous solution was added to neutralize the HCl and the BET was precipitated from the mixture. The precipitate was collected through suction filtration and rinsed with deionized water to afford a white powder (10.35 g, 95.5%). ¹H NMR (600 MHz, DMSO-*d*₆, ppm, Fig. S1): 12.22 (s, -NH), 7.43–7.50, 7.08–7.15 (m, C₆H₄), 3.07–3.12 (m, -CH₂CH₂SH), 2.91–2.98 (m, -CH₂CH₂SH). FTIR (Fig. S2): 2560 cm⁻¹ (SH), 1540 cm⁻¹ (N-H), 1623 cm⁻¹ (C=N).

2.3. Synthesis of polydimethylvinylsiloxane (PVMS)

Teramethylammonium hydroxide in methanol (1.53 g) was added as catalyst to a three-neck flask, and methanol was removed under vacuum afterwards. The catalyst concentration was kept at 0.5% on a monomer basis. Then D4 (11.4 g) and D4^{Vi} (3.96 g) were added to the flask, and the mixture was stirred for 0.5 h at 90 °C under argon atmosphere. MM (0.83 g) was added to the flask, stirred for another 5 h, and heated to 150 °C to decompose the catalyst. The reaction mixture was precipitated in excess methanol and dried under vacuum for 24 h. The PVMS was obtained as colorless viscous liquid. ¹H NMR (600 MHz, CDCl₃, ppm, Fig. S3): 0.04–0.18 (m, -SiCH₃), 5.77–6.05 (m, -SiCH=CH₂). FTIR (Fig. 1b): 3060 cm⁻¹ (C=C-H), 1600 cm⁻¹ (C=C), 1020 cm⁻¹ (Si-O-Si). The number average molecular weight is 5700 and polydispersity (PDI) is 1.64. The molar content of vinylsiloxane unit is 20%, which is determined by ¹H NMR.

2.4. Tethering benzimidazole functionalities onto the PVMS copolymer (PVMS-BET)

PVMS-BET copolymer with different BET contents was synthesized by thiol-ene click reaction (Scheme 1c). The feeding ratio is shown in Table 1. Typically, PVMS (5.7 g, 1 mmol, average of 16 eq. vinyl groups per mol), BET (3.42 g, 19 mmol, thiol/vinyl = 1.2/1) and 0.114 g DMPA (2 wt% of PVMS) were dissolved in a glass vessel with 15 mL THF/DMF (v/v = 2/1) solvent. The vessel was placed under UV light irradiation and stirred for 1 h. Then the solvent was removed by rotary evaporator and dried under vacuum for 24 h after precipitated in methanol. ¹H NMR (600 MHz, CDCl₃, ppm, Fig. S4): 0.04–0.18 (m, -SiCH₃), 0.85 (s, -SiCH₂CH₂), 2.61 (m, -SiCH₂CH₂SCH₂), 3.01 (m, -SiCH₂CH₂SCH₂CH₂), 3.21 (m, -CH₂SCH₂CH₂), 7.43–7.50, 7.08–7.15 (m, C₆H₄). The PVMS-BET copolymer was designated as B-x, where B and x refer to BET and the molar percentage of BET-containing units over total structure units of the copolymer, respectively. The x could be determined as this formula as following:

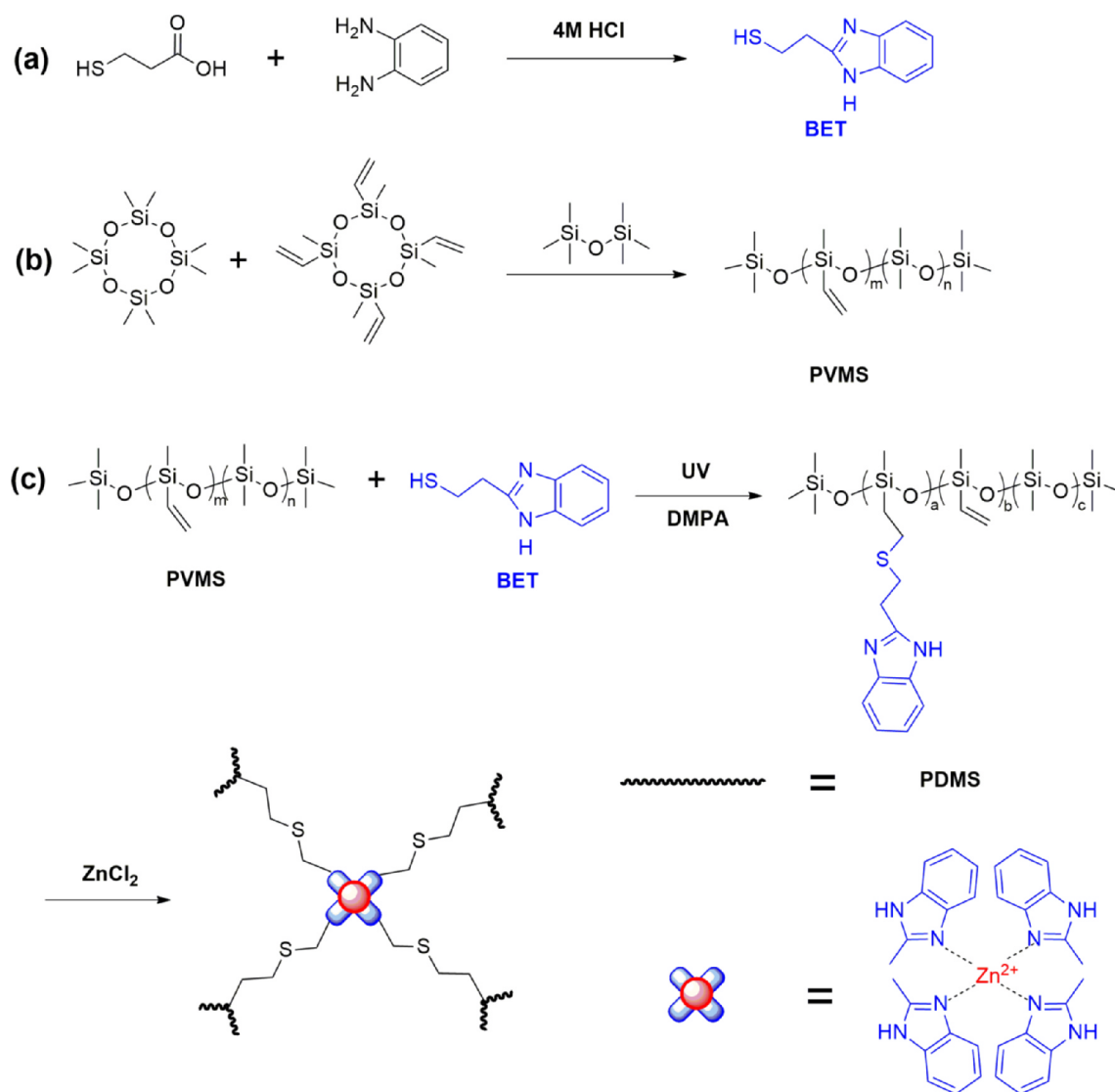
$$x = a/(a + b + c) \times 100$$

2.5. Synthesis of PVMS-BET-Zn polymer

PVMS-BET-Zn polymer was prepared by coordination reaction between ZnCl₂ and B-x. The typical procedure is as follows. ZnCl₂ in THF was added to a solution of B-x in THF under stirring. The BET content was calculated from ¹H NMR, and the molar ratio of ZnCl₂ and BET is 0.25. The mixture was stirred for 12 h at room temperature. Then the mixture solution was casted on different substrates to form polymer film. PVMS-BET-Zn polymer was designated as BZn-x for convenience.

2.6. Characterization

Nuclear magnetic resonance spectroscopy (¹H NMR) was recorded on a Bruker AV600 NMR spectrometer at 25 °C using CDCl₃ or DMSO-*d*₆ as the solvent and tetramethylsilane (TMS) as the interior label. Gel permeation chromatography (GPC) was carried out on Agilent 1260 Infinity with refractive index (RI) and ultraviolet (UV) detectors. The



eluent was THF (flow rate: 1.0 mL/min), and the test was performed at 35 °C with two identical PLgel columns (5 μ m, MIXED-C). Calibration curve was determined by pretests on polystyrene standards (1.30×10^3

to 2.21×10^6 g/mol) with narrow polydispersity. Fourier transform infrared spectroscopy (FTIR) was recorded on a Bruker spectrometer (VECTOR-22 IR) at a resolution of 4 cm^{-1} . Differential scanning

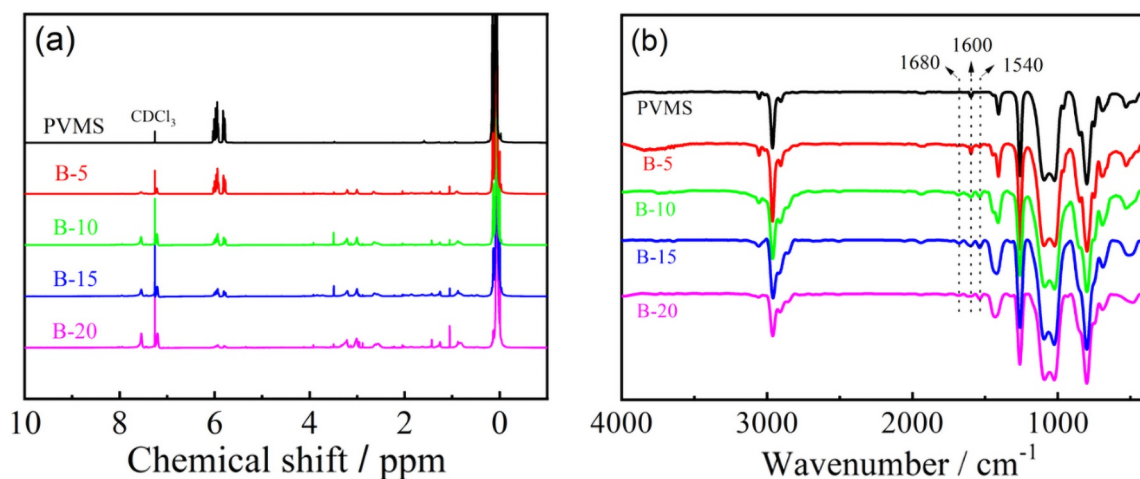


Fig. 1. ^1H NMR (a) and FTIR (b) of PVMS, B-5, B-10, B-15, and B-20.

Table 1
Feeding ratio and characterization data of B-x copolymers.

Sample	Feeding ratio (g)		M_n^a (kDa)	D_M^a	Content of BET ^b (%)	M_c^c (g mol ⁻¹)	T_g^d (°C)
	PVMS	BET					
B-5	5.70	0.86	6.2	3.78	5.5	1590	— ^e
B-10	5.70	1.71	6.7	3.88	10.7	1210	−32.9
B-15	5.70	2.56	7.7	3.22	14.5	1020	18.5
B-20	5.70	3.42	8.1	3.18	19.1	820	27.0

^a Determined by GPC. ^b Calculated from ¹H NMR. ^c ^d Samples after cross-linked by ZnCl₂. M_c is the average molecular weight between two cross-linking sites, which is calculated from ¹H NMR. T_g was tested by DSC. ^e The T_g of BZn-5 is lower than the minimum detection range of the instrument and the relative data can be found in ref. [40].

calorimetry (DSC) measurements were conducted on a differential scanning calorimeter (Netzsch DSC 204F1) under a 20 mL/min nitrogen flow. The sample was quickly heated to 150 °C and kept for 5 min to remove the thermal history. Then it was cooled down to −80 °C at a rate of 10 °C/min and equilibrated for 5 min. Subsequently, it was reheated to 150 °C at the same rate. The glass transition temperature (T_g) was obtained from the midpoint of the endothermic shift during the second heating scan. Tensile tests were performed on dog-bone-shaped specimens using a Shimadzu AG-X plus testing system at a stretching rate of 25 mm/min. Three specimens of each material were tested to get an average value.

2.7. Water contact angle (WCA) and surface free energy (SFE)

Static contact angle measurements were conducted on an OCA40 instrument (DataPhysics) by depositing 3 µL water or diiodomethane droplets on a polymer film surface coated on silicon wafer using the sessile method at 25 °C. The average of the results from five measurements was used as the final CA value. The surface free energy was calculated using the Owens-Wendt-Rabel-Kaelble method [41].

2.8. Atomic force microscopy (AFM) measurements

The surface elastic modulus was measured by AFM (Park Systems, Korea) and determined based on the Hertzian contact theory, which correlates the AFM force signal (F) with the indentation depth of the cantilever tip (δ) and elastic modulus (E) by the following equation [42]:

$$F = \frac{E}{1 - \nu^2} \frac{\tan \alpha}{\sqrt{2}} \delta^2$$

where ν is the Poisson's ratio and is 0.5 for all coatings, α is the face angle of the four-sided pyramidal tip of the cantilever (22°), and δ is the indentation depth.

2.9. Adhesion test

A PosiTest AT-A Automatic adhesion tester from DeFelsko was utilized to measure the adhesion of the polymer film which was solution casting on a glass fiber-reinforced epoxy resin substrate according to ASTM D4541. An aluminum dolly with a diameter of 20 mm was adhered to the surface of the film using epoxy adhesive (Araldite). The adhesion strength was gathered by detaching the aluminum dolly from the substrate at a speed of 0.2 MPa/s. An average adhesion value reported was calculated from five different test points on each coating. The surface of PDMS was sanded before test.

2.10. Pseudobarnacle adhesion test

The test was performed following ASTM D5618 (2011). An

aluminum stud (diameter 10 mm, height 10 mm) was glued to the coating surface using an epoxy adhesive (Araldite). The adhesive was cured for 3 days under ambient conditions. The maximal shear force was measured by a force gauge (SUNDOO SH-500).

2.11. Self-healing test

The polymer film was coated on glass. The self-healing test of the polymer film in air or artificial seawater (ASW) at room temperature was recorded using an optical microscope.

2.12. Antibacterial assays

We used two methods to examine the antibacterial performance. The polymer film was coated on silicon wafer. 1) The coated film was immersed in marine bacteria (*Pseudomonas* sp.) suspension of ASW (1 mL, 10⁸ cells/mL) at 30 °C for 5 h. Subsequently, the film was rinsed with ASW and then stained with the LIVE/DEAD BacLight Bacterial Viability Kit. The adhered bacterial cells were examined under a fluorescence microscope (Scope A1, Zeiss). 2) A coated film (2 × 2 cm²) was immersed in a 3 mL bacterial suspension of ASW with a concentration of 10⁵ cells/mL for 5 h. Subsequently, the coated film was gently rinsed with sterile ASW to remove the unattached bacteria and then placed in a tube with 10 mL sterile ASW followed by a 10-min water bath ultrasonication to remove the strongly adhered bacteria. The above solution was diluted 5 times with sterile ASW, and then 100 µL of the diluted solution was spread onto the MB2216 marine broth culture plates, which were incubated at 30 °C for 48 h to give visible colonies. The number of colonies on each plate was calculated, and each experiment was performed in triplicate.

2.13. Anti-diatom test

The diatom *Navicula incerta* (*N. incerta*) was obtained from Xiamen University. It was cultivated in F/2 culture medium under a 12:12 h light-dark cycle at 23 °C in a light incubator. A 1.0 mL sample of diatom suspension (a chlorophyll value of 0.25 µg/L) was added to a 24-well plate with a silicone slide coated the film. After culturing for 24 h, the slide was gently washed with ASW to remove the unattached diatoms. Adhered diatoms were observed under a microscope. The counts were made for ten fields of view on each of three replicate slides, and the average value of the density of the attached diatoms was recorded.

3. Results and discussion

3.1. Synthesis of the BZn-x copolymer

The synthesis of BZn-x copolymer was illustrated in Scheme 1. First, BET was prepared by reaction of 1,2-diaminobenzene with 3-mercaptopropionic acid under acidic conditions. PVMS was prepared by anionic ring-opening polymerization of D4 and D4^{Vi} with end-capping agent (MM). Then, thiol-ene click reaction between PVMS and BET yielded PVMS-BET. Fig. 1a shows the ¹H NMR of PVMS, B-5, B-10, B-15, and B-20. Compared with PVMS, B-x copolymers present the signals of BET and the peak at 5.7 to 6.1 ppm assigned to vinyl groups from B-5 to B-20 gradually disappeared. Fig. 1b shows that as BET content increases, the bands at 1540 and 1680 cm⁻¹ for the imidazole ring appear in PVMS-BET but the band at 1600 cm⁻¹ for C=C gradually disappears, indicating the successful synthesis of PVMS-BET. As regards the cross-linking of PVMS-BET via metal-ligand coordination between benzimidazoly groups and zinc ions, Fig. S5 shows bathochromic shift related to N-H deformation from 1540 to 1531 cm⁻¹ when ZnCl₂ is added to B-10, clearly indicating the formation of Zn²⁺-imidazole complexes [43]. As shown in Table 1, M_c decreases with the increase of Zn²⁺-imidazole complexes, indicating the increase of cross-link density. The T_g of BZn-x increases with the content of Zn²⁺-imidazole

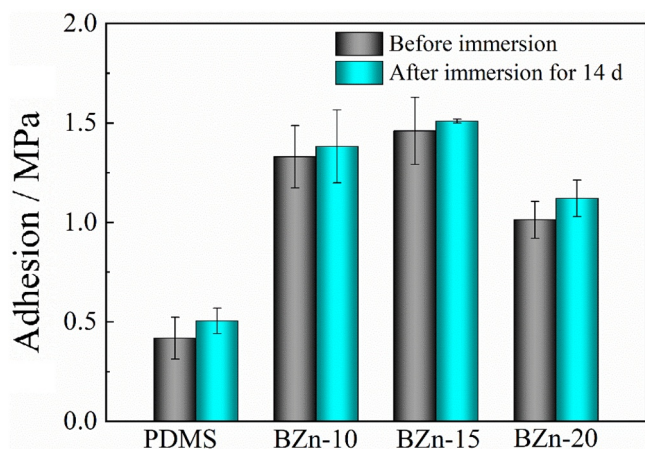


Fig. 2. Adhesion strength of BZn-x coatings before and after immersion in ASW for 14 d.

complexes. This is because the increased cross-link density gradually suppresses the mobility of the polymer chain segments.

3.2. Adhesion strength of BZn-x coatings

Fig. 2 presents the adhesion strength of BZn-x coatings on epoxy panel before and after immersion in ASW for 14 days, where BZn-5 was not used because it was too viscous to form a coating. As expected, PDMS has weak adhesion strength (~ 0.4 MPa) because of its nonpolar structure, which makes it easy to detach from the substrates [44]. However, as the presence of the complex content increases, the adhesion strength generally increases. BZn-10 and BZn-15 have a

significantly higher adhesion (1.3 MPa and 1.4 MPa, respectively), which meets the requirement for marine antifouling coating in applications. On the one hand, the introduction of Zn^{2+} -imidazole complexes significantly enhanced the mechanical properties, which may be the critical factor to improve the adhesion strength. On the other hand, the N-H groups in the imidazole ring can form a strong hydrogen bond with the substrate, improving the adhesion strength. However, further increasing the complex content can decrease the adhesion. BZn-20 has a lower adhesion strength (~ 1.0 MPa) than either BZn-10 or BZn-15. This is understandable because there exists strong hydrogen bonding interactions between the benzimidazole units. The hydrogen binding increases with the BET content. However, after introducing zinc ions, some benzimidazole units coordinated with zinc ions, and thus reducing the hydrogen bonding. Moreover, the coating becomes too brittle to withstand the pull-off testing. After immersion in ASW for 14 days, the BZn-x coatings still have a high adhesion strength similar to that before immersion, indicating all the coatings are stable in marine environments.

3.3. Surface properties and removal strength of pseudobarnacles of BZn-x coatings

Fig. 3a and b show the WCA and SFE of BZn-x coatings. It is known that PDMS has hydrophobic surface with a WCA of $\sim 116^\circ$ and low SFE of $22 \text{ mJ}\cdot\text{m}^{-2}$ due to its nonpolar nature [45]. Each BZn-x coating is also hydrophobic with a similar average WCA around $106\text{--}107^\circ$, indicating that the silicone component is still on the surface. Moreover, the hydrophobicity makes the coating stable during long-term seawater immersion (Fig. S6). On the other hand, BZn-x coatings have a low SFE ($\leq 25 \text{ mJ}\cdot\text{m}^{-2}$) and it slightly increases with BET content. Note that elasticity plays an important role in the FR performance and self-healing of the coatings. Fig. 3c shows the surface elastic modulus (SEM)

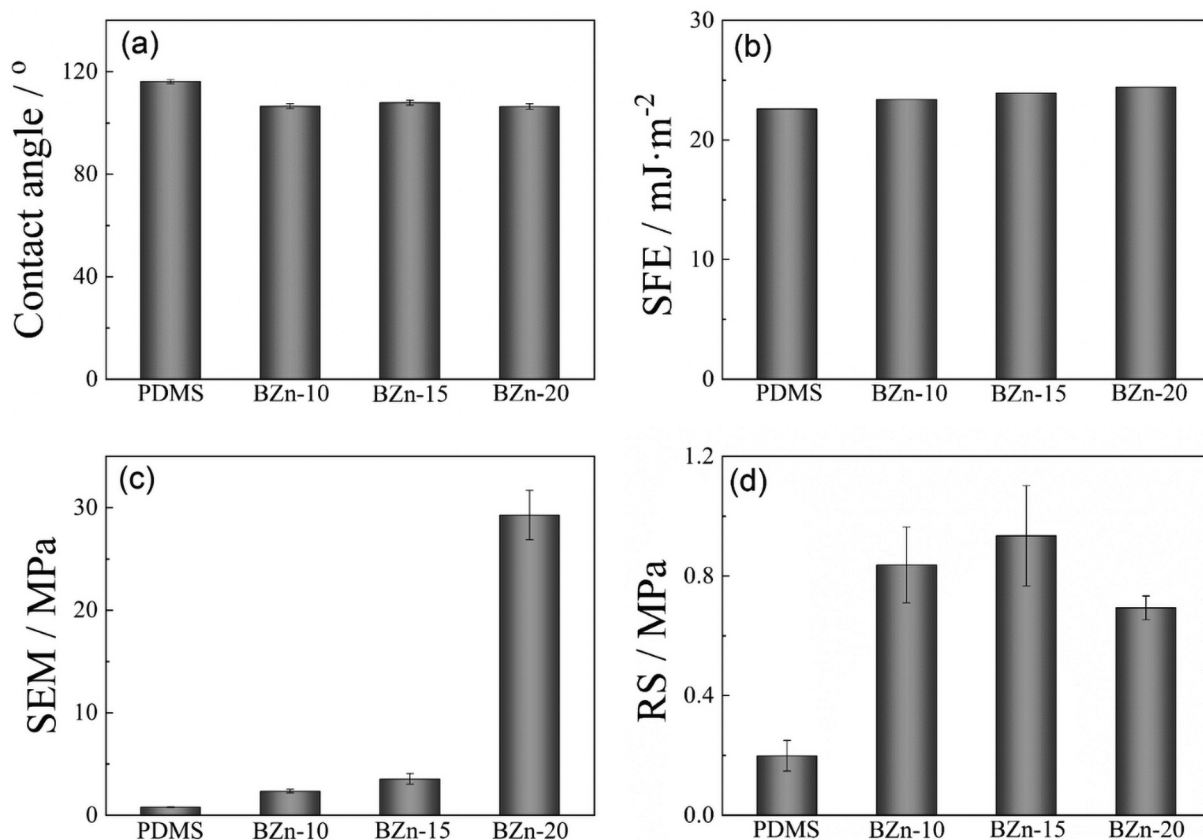


Fig. 3. Static water contact angle (a), surface free energy (SFE) (b), surface elastic modulus (SEM) (c), and removal strength (RS) (d) of pseudobarnacles of BZn-x coatings.

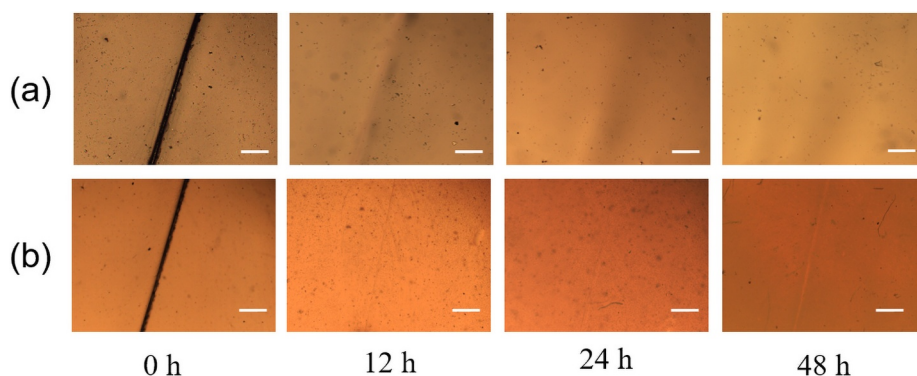


Fig. 4. Self-repairing of sample BZn-10 imaged by an optical microscope: at room temperature in air (a) and immersed in ASW (b). The scale bar is 500 μm .

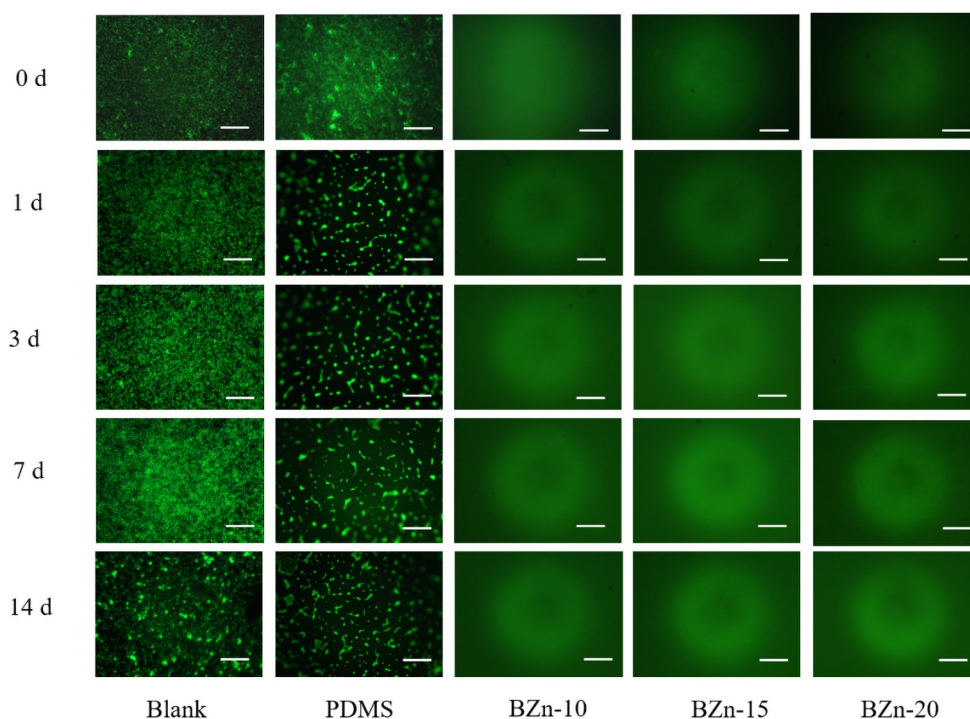


Fig. 5. Fluorescence microscopy images of *Pseudomonas* sp. adhered on BZn-x surfaces after immersion in ASW for 0, 1, 3, 7, and 14 d. The scale bar is 50 μm .

of BZn-x coatings increases with BET content. However, either BZn-10 or BZn-15 has a SEM (2.3 and 3.5 MPa, respectively) slightly higher than PDMS (0.8 MPa), indicating that they still have high enough elasticity. Further increasing BET content leads the coating to lose elasticity. Accordingly, BZn-20 with modulus of 29.2 MPa does not have elasticity in that its high cross-linking density and the steric hindrance of the benzimidazole groups prevent the chain mobility.

Fig. 3d shows the removal strength (RS) of pseudobarnacles of each BZn-x coating, which reflects the fouling release ability. As expected, PDMS has good fouling release with a low removal strength of ~ 0.2 MPa. Each BZn-x coating has a removal strength higher than PDMS, indicating the introduction of Zn^{2+} -imidazole complexes has negative effect on the fouling release performance. As BET content increases, the removal strength generally increases, or the fouling release ability decreases. As discussed above, the decrease in elasticity in the latter is responsible for the decline in their fouling release performance. On the other hand, the formation of coordination bonds or hydrogen bonds between coatings and the aluminum studs also contributes to the higher removal strength. Note that the fouling release performance of BZn-20 is somewhat better than BZn-10 and BZn-15. This is probably because BZn-20 with high modulus is a plastic rather than an elastomer

[46].

3.4. Self-healing properties of BZn-x coatings

The self-healing property of BZn-x film was examined by using an optical microscopy (Fig. 4) and the mechanical properties were tested by a tensile test (Fig. S8). PDMS elastomer is unable to self-repair as their cross-linking via covalent bonds is irreversible [47]. The sample was cut with a razor blade and placed at room temperature without any further treatment. BZn-10 exhibits excellent self-healing performance. The crack in BZn-10 film was gradually healed and disappeared after 48 h at room temperature regardless of whether it was placed in the air or immersed in ASW. Clearly, Zn^{2+} -imidazole coordination bonds between polymer chains are responsible for the self-repairing. They can be broken by physical action upon damaging. Due to the chain mobility and low energy for the binding of imidazole with Zn^{2+} , the dissociated imidazole can readily find the free zinc ions and form coordination bonding again after a certain period of time, making the self-healing reversible [28,48]. However, BZn-10 with a low strength cannot be detected by the testing system. BZn-15 cannot completely self-heal after 48 h (Fig. S7). The original BZn-15 exhibits an ultimate strength of

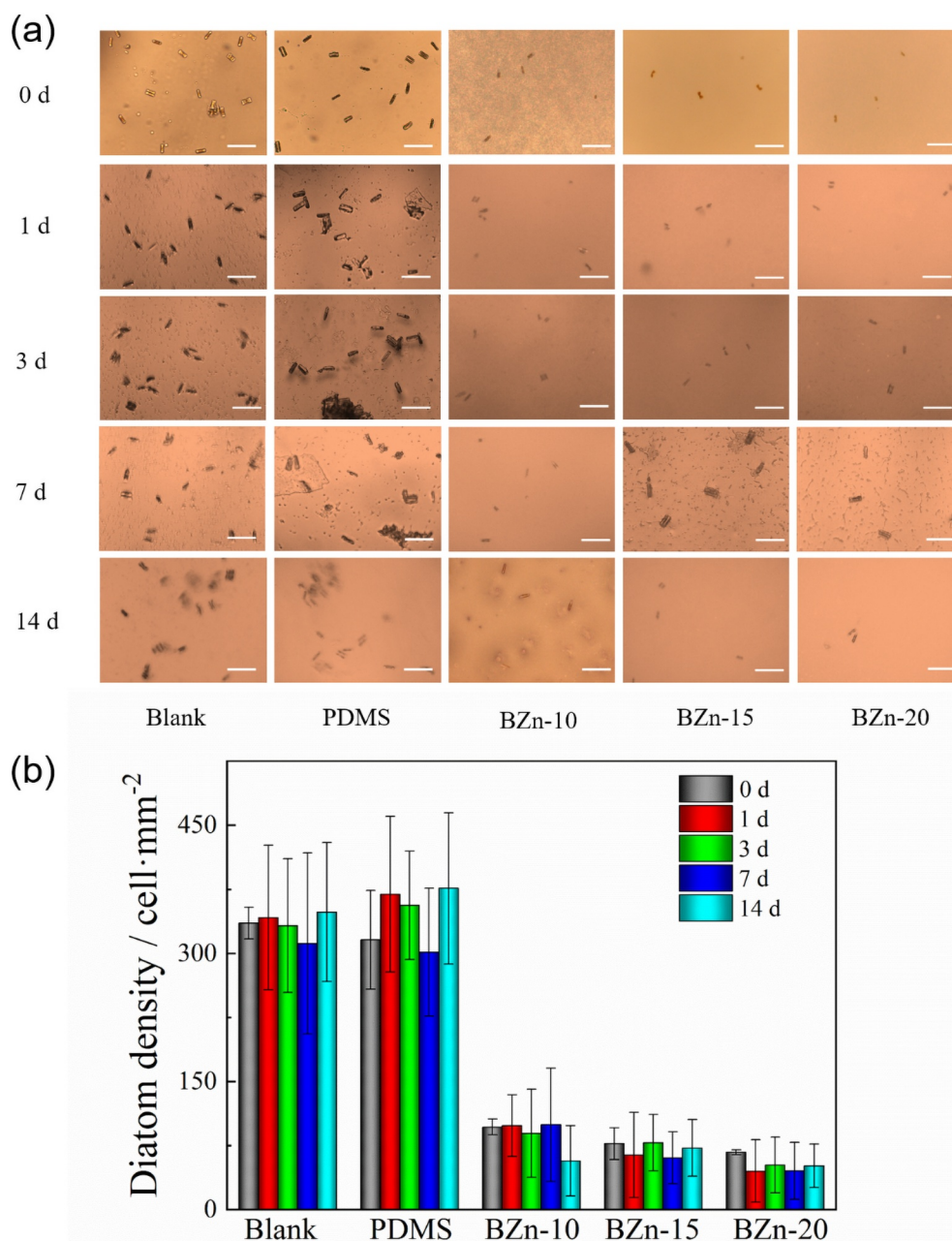


Fig. 6. Optical microscopy images of the diatom *N. incerta* adhered on BZn-*x* coatings after immersion in ASW for 0, 1, 3, 7, and 14 d (a) and the diatom density (b). The scale bar is 50 μ m.

119 kPa and an elongation of 723%. After 48 h of self-repairing, the ultimate strength and elongation restored to 59 kPa and 553%, respectively. BZn-20 has a high ultimate strength of 743 kPa and an elongation of 300%, but it cannot self-repair due to its high cross-linking density which greatly constrains the mobility of polymer chains (Figs. S7 and S8). As BET content increases, the self-healing ability decreases. In other words, the self-healing ability becomes poorer as the polymer elasticity decreases. This is understandable because the cross-linking density increases with Zn^{2+} -imidazole complexes and the chain mobility decreases.

3.5. Antibacterial assays of BZn-*x* coatings

Benzimidazole derivatives have a broad range of biological functions and pharmacological applications, and they have been used as antifungal agents [49,50]. We examined the antibacterial performance

of BZn-*x* coatings by using fluorescence staining method for antibacterial assay after exposure to ASW suspension of marine bacteria *Pseudomonas* sp. for 5 h. As shown in Fig. 5, a number of bacterial cells adhered on the blank silicon wafer (the control) and PDMS control surface. Namely, they were highly subject to bacteria adhesion. On the other hand, almost no bacterial cells were observed on the surface of BZn-*x* coatings, indicating their good fouling resistant performance. We also examined the antibacterial performance of BZn-*x* coatings after immersion in ASW for 1, 3, 7, and 14 d. The coatings still showed excellent antibacterial activity, indicating their stability of fouling resistance.

The antibacterial activity was also examined with respect to the colony forming units (Fig. S9). The bacterial viability was calculated by the bacteria adhered on the coating using the blank sample as the control. PDMS has poor fouling resistance, which subjected to bacterial colonies with bacterial viability of 93%. In contrast, the antibacterial

rate for each BZn-x coating against *Pseudomonas* sp. is more than 96%. The antibacterial ability increases with BET content, indicating the Zn^{2+} -imidazole complexes have antifouling function [51]. This is consistent with the fluorescence staining results. Accordingly, BZn-x coatings have fouling resistant ability.

3.6. Anti-diatom performance of BZn-x coatings

Fig. 6 shows the optical images of diatom *N. incerta* adhered to BZn-x coatings with its density after immersion in ASW for 0, 1, 3, 7, and 14 d. A great number of diatoms have colonized on PDMS surfaces ($\sim 320 \text{ cell}\cdot\text{mm}^{-2}$) in that they like hydrophobic surfaces [52]. In contrast, the diatom density on BZn-x coatings is markedly lower whether they are immersed in ASW or not. The density before immersion in ASW is about 96, 77 and 67 $\text{cell}\cdot\text{mm}^{-2}$ on BZn-10, BZn-15, and BZn-20 surfaces, respectively. Namely, the anti-diatom ability increases with BET content. The similar anti-diatom performance of BZn-x coatings after immersion in ASW for 1, 3, 7, and 14 d indicates the stability of fouling resistance. As stated above, the Zn^{2+} -imidazole complexes have fouling resistance function acting as non-leaching antifoulants. It is favorable to improve fouling resistance though it reduces the fouling release performance. Optimization of the composition of the polymer is critical for the properties of the coating.

4. Conclusions

We have developed silicone based fouling-resistant coating with self-healing performance. The coating has hydrophobic surfaces with low surface energy. It has improved adhesion strength to the substrate in comparison with PDMS due to the hydrogen bonding. The self-healing ability arises from the coordination between Zn^{2+} and imidazole groups attached on PDMS chains. The Zn^{2+} -imidazole complexes also serve as non-leaching antifoulants, allowing the coating to exhibit good fouling resistant performance toward bacterial (*Pseudomonas* sp.) and diatom (*N. incerta*) adhesions. Moreover, the fouling resistance is stable with the immersion time. Compared with BZn-10 and BZn-20, BZn-15 shows good self-healing property under room temperature with moderate mechanical properties and excellent fouling resistance. This coating with fouling resistance and self-healing is expected to find application in marine antifouling.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2020.126870>.

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